

Problem Statement 29 Municipal Infrastructure Damage Reporting App

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Section: C

Domain: Smart Cities, Transport & Logistics

Actors: Citizen Reporter, Municipal Engineer

Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional – Geolocation	The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint the incident location on the municipal GIS map.	High	Pass: Pothole report routed to Road Infrastructure division with exact GPS coordinates. Fail: Report submitted without location data.	Location accuracy is the single most important input for correct department dispatch and field-crew navigation.
FR-002	Functional – Data Capture	The system shall allow a Citizen Reporter to submit a damage report by selecting a category (Pothole, Streetlight, Water Leak), attaching	High	Pass: Report is saved with category, photo(s), timestamp, and citizen ID	Structured categorisation is required before automated routing (FR-

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		up to 3 photos, and adding an optional text description.		attached. Fail: Report accepted without a selected category.	003) can function.
FR-003	Functional – Workflow / Routing	The system shall automatically dispatch a newly created ticket to the correct municipal engineering division based on the report category and GPS zone, and generate a unique ticket ID.	High	Pass: A “Streetlight” report is auto-assigned to the Electrical Maintenance division within 60 seconds of submission. Fail: Ticket remains unassigned after 5 minutes.	Manual triage causes delays; automated routing is the core value proposition of the app.
FR-004	Functional – Tracking / Notification	The system shall allow a Citizen Reporter to view real-time ticket status (Submitted → Acknowledged → In Progress → Resolved) and receive a push/SMS notification on every status change.	Medium	Pass: Citizen receives a notification within 2 minutes of the Municipal Engineer updating ticket status. Fail: Status change occurs with no citizen-facing notification.	Transparency increases citizen trust and reduces duplicate/repeat reporting of the same issue.

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FR-005	Functional – Ticket Management	The system shall provide a Municipal Engineer dashboard to view all tickets assigned to their division, filter by status/priority/location, update ticket status, and upload resolution proof photos before closing a ticket.	Medium	Pass: Engineer closes a ticket only after attaching at least one “after” photo. Fail: Ticket can be marked Resolved with zero attachments.	Photographic proof of resolution is required for municipal audit and citizen accountability.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The reporting portal shall support offline report drafting and auto-sync when mobile network connectivity resumes; all synced data shall be transmitted over TLS 1.2+.	High	Pass: Benchmarking tests confirm target latency (sync completes within 10s of reconnect) and TLS encryption on all API calls under simulated peak load. Fail: Draft data lost on connectivity drop, or data transmitted unencrypted.	Field connectivity in low-infrastructure areas (where damage is most common) is unreliable, and location/photo data is sensitive.
NFR-002	Usability & Accessibility	The citizen-facing app shall support at least 2 regional languages (in addition to English) and provide voice-assisted report submission for low-literacy users.	Medium	Pass: A user can complete a full report submission using only voice prompts in the selected regional language. Fail: App is unusable without English text input.	Municipal citizen bases are linguistically diverse; accessibility directly affects civic-tech adoption rates.